Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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1. Let

$$A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Compute A^2, A^3, A^4, and A^5.

Solution:

$$A^2 = A \cdot A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Since $A^2 = A$, it's clearer that, by the definition of the matrix power:

$$\forall p \in \mathbb{N}^+, \quad A^p = A^{p-1} \cdot A \implies$$

Therefore:

$$\begin{aligned} \implies A^3 &= \underbrace{A^2}_{A} \cdot A = A \cdot A = A^2 = A \\ \implies A^4 &= \underbrace{A^3}_{A} \cdot A = A \cdot A = A^2 = A \\ \implies A^5 &= \underbrace{A^4}_{A} \cdot A = A \cdot A = A^2 = A \end{aligned}$$

Alternative solution: (extra)

Since

$$\boxed{A^2 = A}$$

we may avoid the efford of calculte the other powers using induction.

This means:

$$\boxed{\forall n \in \mathbb{N}^+, \quad A^n = A}$$

Base case ($n = 1$):

$$A^1 = A.$$

Inductive hypothesis:

Let $n \in \mathbb{N}_{>0}$ and assume that $A^n = A$.

Inductive step:

Multiplying both sides of the inductive hypothesis by A on the right:

$$A^n = A \implies A^n A = AA.$$

Therefore,

$$\begin{aligned} A^{n+1} &= A^n A \\ &= AA \\ &= A^2 \\ &= A. \end{aligned}$$

Therefore, $P(n) \implies P(n+1) \implies \boxed{A^n = A}$

2. Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}.$$

Verify that there exists a power of A which is the zero matrix.

Solution:

Basically, they are asking for:

$$A^x = 0 \in \mathcal{M}_{0\times 0} \tag{1}$$

Given $A \in \mathcal{M}_3(\mathbb{R})$, we must verify that

$$\exists k \in \mathbb{N}_{>0} : A^k = 0_3.$$

Computing A^2:

$$
\begin{aligned}
A^2 &= AA \\
&= \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \neq 0_3.
\end{aligned}
$$

Computing A^3:

$$
\begin{aligned}
A^3 &= A^2 A \\
&= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \\
&= \boxed{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}.
\end{aligned}
$$

Therefore,

$$
\boxed{\exists k = 3 \in \mathbb{N}_{>0} : A^k = 0_3}.
$$

3. Consider the matrices

$$
A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.
$$

Compute $(AB)^2$, A^2B^2, and B^2A^2.

Solution:

Given $A, B \in \mathcal{M}_{2\times 2}(\mathbb{R})$, all three products are defined.

Product $(AB)^2$:

First, compute AB:

$$
\begin{aligned}
AB &= \begin{bmatrix} 1(1) + 1(-1) & 1(1) + 1(1) \\ 1(1) + 0(-1) & 1(1) + 0(1) \end{bmatrix} \\
&= \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix}.
\end{aligned}
$$

Therefore,

$$(AB)^2 = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}}.$$

Product A^2B^2:

Computing both squares:

$$A^2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix},$$

$$B^2 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}.$$

Then,

$$A^2B^2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} -2 & 4 \\ -2 & 2 \end{bmatrix}}.$$

Product B^2A^2:

$$B^2A^2 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} 2 & 2 \\ -4 & -2 \end{bmatrix}}.$$

Hence,

$$\boxed{(AB)^2 \neq A^2B^2 \neq B^2A^2}.$$

In general, $(AB)^2 = ABAB$, which cannot be rearranged as A^2B^2 without assuming $AB = BA$.

4. State, with justification, whether the following statements are true or false.

   a) For any square matrices A and B of the same order, $AB = BA$.

   b) For any square matrices A and B of the same order, if $AB = 0$, then $A = 0$ or $B = 0$.

   c) For any square matrices A, B, and C of the same order, if $AB = AC$, then $B = C$.

   d) If, for a square matrix A, there exists $k \in \mathbb{N}$ such that $A^k = 0$, then $A = 0$.

   e) For any square matrices A and B of the same order, $(AB)^2 = A^2B^2$.

5. Final item. Remove when done.